SeEar: Tailoring Real-time AR Caption Interfaces for Deaf and Hard-of-Hearing (DHH) Students in Specialized Educational Settings

Yasith Samaradivakara
2019is074@stu.ucsc.cmb.ac.lk
Mixed Reality Lab, University of Colombo School of Computing
Sri Lanka
Augmented Human Lab, National University of Singapore
Singapore

Thavindu Ushan
2019is050@stu.ucsc.cmb.ac.lk
Mixed Reality Lab, University of Colombo School of Computing
Sri Lanka

Asela Pathirage
2019is053@stu.ucsc.cmb.ac.lk
Mixed Reality Lab, University of Colombo School of Computing
Sri Lanka

Prasanth Sasikumar
prasanth@ahlab.org
Augmented Human Lab, National University of Singapore
Singapore

Kasun Karunanayaka
ktk@ucsc.cmb.ac.lk
Mixed Reality Lab, University of Colombo School of Computing
Sri Lanka

Chamath Keppitiyagama
chamath@ucsc.cmb.ac.lk
University of Colombo School of Computing
Sri Lanka

Suranga Nanayakkara
suranga@ahlab.org
Augmented Human Lab, National University of Singapore
Singapore

Figure 1: Initial Prototype to Real-World Application through User-Centered Design

ABSTRACT

Deaf and hard-of-hearing (DHH) students often encounter challenges in their specialized educational settings, particularly in language literacy development, due to limited exposure to written and spoken language. We, therefore, collaborated with 8 DHH students and 2 Teachers of the Deaf (ToDs) from a School for Deaf in Sri Lanka to explore how to effectively utilize a real-time augmented reality (AR) caption interface in their classroom settings, aiming to enhance their learning experience. We adopted a User-Centered Design (UCD) process to develop and refine the prototype addressing the design challenges affecting the overall learning experience. User study with 12 DHH participants revealed a strong preference (91.66%) for our prototype and highlighted its potential to enhance the learning experiences. Our findings contribute to further exploration of the potential of tailored AR caption interfaces in addressing the educational challenges faced by DHH students and enhancing their learning experience in specialized educational settings.

CHI EA '24, May 11–16, 2024, Honolulu, HI, USA

ACM ISBN 979-8-4007-0331-7/24/05
https://doi.org/10.1145/3613905.3650974

CCS CONCEPTS

• **Human-centered computing** → **User centered design**; **Accessibility**; • **Applied computing** → Education.

KEYWORDS

Augmented Reality, Deaf, Real-time Captioning, Co-Deign, Information Interfaces and Presentation

ACM Reference Format:

Yasith Samaradivakara, Thavindu Ushan, Asela Pathirage, Prasanth Sasikumar, Kasun Karunanayaka, Chamath Keppitiyagama, and Suranga Nanayakkara. 2024. SeEar: Tailoring Real-time AR Caption Interfaces for Deaf and Hard-of-Hearing (DHH) Students in Specialized Educational Settings. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24), May 11–16, 2024, Honolulu, HI, USA.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3613905.3650974

1 INTRODUCTION

DHH students are often admitted to specialized schools for the deaf due to the limited availability of resources, trained personnel, and necessary accommodations in mainstream educational settings. In specialized educational settings, students mostly rely on sign language, which limits their exposure to written and spoken language. This limited exposure affects the language literacy development and causes numerous challenges in their social, professional, and educational lives [23, 30, 44].

With the recent advancements in speech recognition, caption interfaces have been used as a promising solution to provide real-time access to spoken words for DHH individuals [9, 29, 36]. However, in the educational context, most of these caption interfaces are displayed either through projectors, students' laptops, or mobile phones. These methods present challenges because students must constantly shift their attention between the instructor and the caption interface, which significantly increases their cognitive load [21]. With recent developments, these caption interfaces have evolved from traditional digital displays to immersive AR interfaces, offering the potential to deliver information seamlessly within the user's field of view which enhances glanceability, enabling visual contact with speakers, and allowing for a hands-free experience.

Previous studies have demonstrated real-time AR caption interfaces as supporting tools for DHH individuals in education (e.g., classroom discussions, lectures) [17, 18]. Extending these findings, we are exploring the application of real-time AR caption interfaces in specialized educational settings, to enhance learning experiences, support educational activities, and improve language literacy. We adopted a UCD process, collaborating with 8 DHH students and 2 Specialized Teachers from a Deaf School in Sri Lanka. We began with a comprehensive thematic analysis to identify the main challenges in current educational settings. Based on these insights, we developed our initial prototype — an AR glass integrated with a smartphone, mounted on low-cost transparent headsets reflecting the smartphone screen into the user's field of view, incorporated with localized real-time captions. Through semi-structured interviews, we identified challenges in the initial prototype, including caption presentation method, placements, and transcription inaccuracies, affecting the overall learning experience. Employing an iterative co-design process and leveraging insights from prior work, we designed prototypes to address these challenges.

We evaluated our prototypes through user studies with 12 DHH students and the results suggested that our implementation of caption presentation methods, and caption placements enhances the learning experience. Strong preference (91.66%) and qualitative feedback from participants suggested that our system enhances the learning experience of students, supports educational activities, and subsequently improves the language literacy of students.

2 RELATED WORK

AR in Education for DHH students: Previous studies have demonstrated the utility of AR tools to support and enhance the learning experience for DHH students in educational settings using visual-centric information (eg. sign avatars) [1, 20, 26]. These studies have shown that such approaches are highly effective in terms of educational enhancements, student interest, active engagement, and collaboration [2, 3, 14]. In our work, we design an AR caption interface to provide an enhanced learning experience, with greater student engagement.

Augmented Reality Caption Interfaces for DHH Education: The development of AR caption interfaces tailored for DHH students in classroom environments poses unique challenges. Marschark *et al.* [27] observed that DHH students relying on real-time captions often underperform compared to their hearing peers. In response, Jain *et al.* [18] explored the use of AR headsets to deliver the captions directly in users' line-of-sight. We build upon their findings, aiming to adapt and refine these interfaces and address identified limitations.

AR Caption Presentation Methods: The impact of text presentation methods on DHH individuals, who have special needs, is a crucial area of exploration. Studies like those of Duchnicky and Kolers [11] and more recent works [35] have delved into the reading speed, comprehension, and task load of non-deaf users with different text presentation methods. However, the impact on DHH individuals who have distinct language proficiency, visual attention, and information processing [12, 25, 33] needs to be further explored. Masson *et al.* [28] identified reading longer text could overload users, leading to reduced comprehension. Moreover, Jain *et al.* [18] identified that presenting captions as single-line at a time affects the comprehension of DHH users, while Peng *et al.* [31] found that adding more lines increases comprehension in communication. In our work, we employ these methods and design more effective presentation methods that enhance the learning experience of students, considering the amount and presentation of text to optimize comprehension, reading speed, cognitive load, and engagement.

AR Caption Placement: The strategic placement of captions in AR environments is crucial for enhancing the learning experience of DHH students. Dynamic captioning, a technique widely used in video captioning for movies and television [8, 16], presents a foundational concept. Jain *et al.* [18] proposed an approach, that allows users to manipulate captions in a 3D space, with a fixed captioning strategy for more static, non-interactive sessions. Our research builds upon these ideas, exploring how different caption placement strategies in AR can influence DHH students' learning experience, and aims to identify optimal solutions while addressing current limitations.

Automatic Speech Recognition (ASR) Transcription Errors: Real-time caption interfaces use ASR as captioning service. However, ASR is imperfect; causing challenges on understandability of captions [19], especially for DHH individuals. Previous studies have explored different approaches by visually representing the transcription errors [5, 37]. Berke *et al.* [5] changed the appearance of text to indicate low-confidence words in video captions. We used these concepts to guide our co-design process.

3 NEED ANALYSIS

Our need analysis was conducted with 8 DHH students (4 female & 4 male, Age: 14-18, Hearing Loss: Moderate - 1, Severe - 4, Extremely Severe - 3) and 2 Teachers of the Deaf (ToD) from a school of deaf in Sri Lanka. These students have been enrolled in the school since the age of 6 and have exclusively received instructions in sign language. The objective was to understand the challenges faced by these students within their educational settings, their adaptation strategies, and the broader impacts of current educational approaches. We conducted focus group discussions and semi-structured interviews to gather data and a thematic approach [6] to analyze the data based on inductive and deductive themes.

Literacy Challenges: Both teachers emphasized the difficulties DHH students face in reading and comprehending written text. For example, one teacher (T2) highlighted students' struggles with paper-based exams, *"Students take a lot of time to read and comprehend exam papers, often struggling to finish within the allocated time."*. This was linked to their limited exposure to written and spoken language in classroom sessions. Some students expressed their struggles with reading speed and understanding written text, with one stating, *"I can read words but it takes more time for me to read and comprehend."* (S4). Unfamiliarity with textbook vocabulary was a common issue, as expressed by another student: *"I find it difficult to read and understand textbooks, as most words are unfamiliar for me."* (S6). Students also mentioned needing additional support, *"It's hard for me to read books alone; I need support with lip reading."* (S7).

Educational Methodology Limitations: The teachers noted the shortcomings of current educational methods in supporting the development of reading and writing skills. One teacher remarked, *"Some words, especially in textbooks, can't be represented in signs. Students find those words hard to read and understand."* (T1). The lack of sign language representation for certain concepts was mentioned as leading to oversimplified content, impacting vocabulary development.

Feedback from DHH students and teachers highlighted the shortcomings of current educational methods (eg. limited exposure to language, sign language limitations) and specific need for tools to develop language proficiency, including reading and comprehension, vocabulary, and correlation with general exam performance.

4 ITERATIVE DESIGN

The core goal of our design process was to provide an optimal AR caption interface for DHH students, effectively utilized in their classroom settings, addressing their unique needs and enhancing their learning experience. To accomplish this, we introduced a preliminary AR interface (Figure 1), similar to traditional captions, displaying multiple lines of text directly in the users' line-of-sight. We then conducted semi-structured interviews to gather feedback about the challenges in the prototype affecting their learning experience. This feedback, combined with active co-design involving DHH students who provided ideal designs and comments, guided our refinements. Finally, we integrated these insights with prior research and user preferences to develop prototypes specifically designed to overcome the identified challenges.

4.1 Semi-Structured Interviews and Co-Design

We conducted a semi-structured interview and co-design with the same 8 DHH students involved in the need analysis. We based our semi-structured interview format on prior work by Hong *et al.* [16]. For each participant, we asked them about the challenges in a prototype that impact their learning outcomes and experiences and their ideal visualization solutions.

Challenges.

(1) **Low Comprehension:** Participants found the amount of text presented at a time overwhelming, making it difficult to grasp the overall meaning and requiring significant focus. P5 highlighted, *"Too many words shown at a time, it's hard to grasp the overall meaning."*. P4 added, *"It's difficult to follow and understand the captions."*. P7 mentioned an issue with placement, *" It's difficult to lip-read and text read due to the text placement and I often miss some text."*
(2) **Low Reading Speed:** Majority (n=5) found reading with the prototype time-consuming. P3 explained, *"It's harder for me to focus on which word to read with a lot of text."*. P6 added that *"Continuing to read further down the caption lines slows me down."*. Participants also mentioned that placement affects their reading speed, *"Consistently shifting my eyes between lip-reading and text reading slows me down."* (P4).
(3) **High Task Load:** Many participants (n=6) felt that the prototype causes excessive mental strain, often leading to discouragement and reluctance to read. For example, P2 mentioned, *"It makes me nervous when there are a lot of words at once."*. P7 stated, *"Lot of text is overwhelming as I feel like I have a big task to complete."*. Additionally, participants suggested that text should be closer to the lip, *"I need to lip read with the words but in here I have to move my eyes which takes a lot of effort."* (P8).
(4) **Transcription Errors:** Participants noticed issues with the transcriptions, P1 mentioned, *"In the middle of a sentence, it showed just a letter. I'm curious whether it showed the actual word the teacher said. However, it distracted the reading."*. P3 also suggested, *'It would be better if you could mark any errors, so I won't be confused.'"*

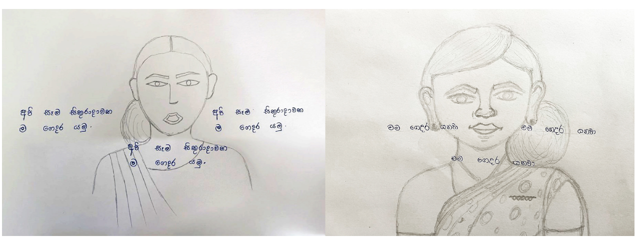
Ideal Designs. After the semi-structured interviews, we asked participants to describe and provide their ideal designs (Figure 2) addressing the challenges identified.

Amount of Content: Most participants (n=6) responded with designs (Figure 2a) maximum of two lines of captions with three words per line. All the participants agreed that the amount of content should not exceed that limit, which makes them discouraged and reluctant to read.

Association with the Speaker: Similar to Jain *et al.*'s study [18] participants liked when captions move with head movement. However, most participants (n=6) found fixed positioning problematic, causing neck fatigue *"I have to move my head to position captions closer to the speaker, which is quite challenging and uncomfortable over an extended period."* (P4). Everyone responded with designs (Figure 2a) with captions closer to the speaker either left, right, or directly below.

Error Handling: Similar to Berke *et al.*'s study [5] participants found changing text appearance to be distracting. Suggested designs include a red squiggly line underneath the word (P1), an angry emoticon/thumbs down emoticon next to the word (P6), a triangle with an exclamation point icon next to the word (P4), a Strikethrough (P8) (Figure 2b). Most (n=5) expressed a preference for a red squiggly line, noting its familiarity with word processing tools.

Additional Feedback: Participants suggested incorporating a finger-guided reading method for text presentation, drawing from their school experiences, which involves sequentially pointing to each word in a passage to read. Participant (P2) remarked, *"Could you highlight each word as I go along? It simplifies the reading process."*.



(a) Caption Presentation and Placement Methods

(b) Markups for Inaccurate Transcribed Word

Figure 2: Ideal Designs by DHH Students: (a) Illustrating the number of caption lines (one or two), words per line (three), and the placement (left to the speaker, right to the speaker, directly below), (b) Markups for inaccurately transcribed word (a-e)

4.2 Design Goal and Proposal

From the challenges we defined and insights from co-design, we categorized challenges into design challenges and proposed AR prototypes to address these issues. Given that Sinhala [43] is the only proficient language for participants, our prototypes are specifically designed with Sinhala Captions.

Caption Presentation Method. The major need of participants is to present the optimum amount of text at once while increasing the reading speed, and comprehension and minimizing the task load. Considering the prior work, and co-design insights we developed a few presentation methods (Figure 3).

- **Karaoke-Like Scrolling:** During the co-design phase, participants suggested a sequential word-by-word highlighting method enabling easier text navigation. We implemented this method using a self-guided approach, where students can use a Bluetooth mouse to navigate sequentially through the words. They preferred mouse-click navigation over visual cursors, which were found to be effort-intensive and distracting.
- **Single-Line Scrolling:** This method splits the captions into lines that present one after the other. Previous studies [35] have proven that it requires small space minimizing cognitive strain and enhancing reading speed.
- **Multi-Line Scrolling:** Similar to Single-Line scrolling, this method splits captions into two lines and is presented at once. Previous studies [31] suggested that two lines improve comprehension than single-line.
- **Rapid Serial Visual Presentations (RSVP):** Introduced by Forster [13], involves presenting text word by word in a rapid sequence at a fixed location, with a red letter appears as a resting point for the user's eyes while reading. This method requires small space and efficient for speed reading.

Highlighting Strategy: To incorporate word-by-word highlighting for the Karaoke-Like presentation method, we chose a few popular text-highlighting strategies based on prior work, Font Color (color_r) [32], Bold Face [34], Background (color_bg) [10], Underlining (ul) [41], Italicized (it) [4] and Font Size (size) [42] (Figure 3).

Placement of Caption: In co-design (Figure 3) participants suggested placing captions in different positions relative to the speaker. Consequently, we developed a prototype that provides the flexibility to reposition and resize captions.

5 EVALUATION

The studies were conducted in accordance with the ethical research guidelines provided by the Ethics Review Committee (ERC) of the University of Colombo School of Computing and with ERC approval.

5.1 Participants

We recruited 12 DHH students (5 female & 7 female, Ages: 12-18, Hearing Loss: Moderate - 2, Severe - 6, Extremely Severe - 4) from the same deaf school for evaluations. Their only proficient language was Sinhala and all had non-disabled vision for daily activities without glasses. Communication was facilitated by a ToD, an expert in Sign Language.

Figure 3: Designed prototypes: a) Caption Presentation Methods, b) Caption Placement Methods, c) Highlighting Strategies

5.2 Study 1: Placement of Captions

5.2.1 Procedure. The Primary objective of this study was to identify the effect of different caption placements on the learning experience. We asked each participant to engage in an individual classroom session which was conducted by the Teacher. Each participant used all four caption placements (Figure 3) and we counterbalanced the order of placements with a Latin square. On average study took about 8 hours in total and spanned three days. Study materials were chosen by the teacher considering participants' proficiency. After the study, each participant provided their qualitative feedback about each caption placement method which we interpreted from the teacher.

5.2.2 Qualitative Findings: Participants (n=7) mentioned reading captions below the teacher/ centered on the screen is easy to understand, *"I can read lips and text at the same time, making it easy to follow and understand"* (P11). While other participants (n=5) find it distracting when captions are positioned in the central view, *"I prefer to keep captions separated from central view, it's easier to switch between teacher and captions without obstructing."* (P6). Participants (n=3) more familiar with book reading wanted positioning captions left on the screen for a natural reading experience, *"I like to keep captions left to the teacher; it gives a more natural reading experience, like reading a book"* (P2). However, all participants agreed that captions should be placed closer to the teacher's face, either left, right, or directly below. Participants who do not rely on lip reading preferred to keep captions either left or right to the speaker, P2 noted, *"When captions are positioned in line with the teacher's face, it feels very comfortable to read, rather than displaying it on the teacher's body."*, P1 *"I don't need to read lips much often, I prefer keeping captions separated from the teacher."*. Different participant feedback suggested that reading speed and task load depend on the preference and unique needs of the user.

Findings suggest the importance of customizable caption placement in enhancing the educational experience. The diverse preferences underscore the need for adaptable captioning systems that cater to individual requirements, thereby optimizing comprehension, comfort, reading speed, and task load in learning environments.

5.3 Study 2: Caption Presentation Method

5.3.1 Procedure: This study was conducted to evaluate the effect of caption presentation methods (Figure 3) on reading speed, comprehension, subjective task load, and overall learning experience. We selected 'directly-below' (Figure 3) as the caption placement, preferred by the majority (58.3%), and Font-Color as the highlighting strategy for Karaoke-Like presentations, preferred by the majority (41.67%). Each participant engaged in four classroom sessions, each from all four presentation methods, a total of $12 \times 4 \times 4 = 192$ assessment sessions. We counterbalanced the order of presentation methods with a Latin-square design. Lesson material was chosen by the teacher from a school textbook [22] chapter and the teacher verbally delivered the lesson. The study spanned six days, averaging 8 hours per day. In Post-study, we calculated reading speed using words per minute (WPM), Subjective Task Load using Raw TLX (RTLX) [15], and comprehension using a comprehension test created by the teacher, graded from 0 to 10. Additionally, participants ranked the most preferred prototype and provided qualitative feedback.

5.3.2 Quantitative Findings: Most participants (n=9) preferred Karaoke-Like presentations over other methods. ANOVA test showed a significance for comprehension among presentation methods ($F(3, 188) = 37.999, p < .05, \eta_p^2 = 0.038$), with paired t-test for post-hoc comparisons revealed higher comprehension for Karaoke-Like over other methods, p < .05. ANOVA for reading speed showed significance ($F(3, 188) = 16.741, p < .05, \eta_p^2 = 0.042$), with post-hoc test, revealed higher reading speed for RSVP, p < .05. Karaoke-Like had a higher reading speed than Multi-Line (p < .01), and Single-Line over Multi-Line, p < .05. Average reading efficiency, calculated by multiplying comprehension and reading speed, was highest for Karaoke-like presentations. RTLX score was lower for Karaoke-Like, without any significance.

5.3.3 Qualitative Findings: Participants found the Karaoke-Like method effective for focusing on one word at a time, enhancing reading effectiveness, *"With the highlighted word, I can focus solely on that, making reading easier."* (P2), *"It's like someone assisting me to read, which I find very effective."* (P4). They perceived it as interactive and motivating, *"I like it very much, I feel like I need to read the words."* (P5), and noted its benefit in reducing mental load by limiting focus to one word. P4 mentioned Karaoke-Like could be further improved using automatic tracking (eg. eye-tracking).

Findings suggest that Karaoke-Like presentations effectively enhance reading efficiency and minimize cognitive strain in daily classroom activities, thereby helping students to engage more effectively for extended periods.

5.4 Overall System Evaluation

Our system (Diagram A.1) comprises two main parts: AR Glass [24] and a Unity [38] Software Development Kit [40] incorporated with prototype features, integrated localized real-time captioning using Microsoft Azure Speech-to-Text [39] and supports multiple platforms. We developed a Unity Sinhala Unicode Asset [40] for Unicode Conversions.

Following the completion of prototype evaluations, participants were invited to share their feedback regarding the overall system. We used the System Usability Scale (SUS) [7] to calculate the overall usability of the system and participants including teachers provided qualitative feedback, analyzed using thematic analysis [6].

5.4.1 Quantitative Findings: The average System Usability Scale (SUS) [7] score of 78.18 (SD=5.23) given by the 12 participants suggests that students found the system to be well-integrated and easy to use. Eleven participants (91.66%) showed their preference to use our system for their classroom activities.

5.4.2 Qualitative Feedback: **Literacy Development and Educational Activity Support:** Students and teachers recognized the system's potential to enhance literacy skills. Teacher T1 highlighted, *"With long-term use of the system students get used to read and comprehend quickly."*. Student S7 mentioned, *"I usually remember words by writing, now I can write when the teacher speaks."*, S10 added, *"Now I can write short notes of the ongoing classroom session."*. T2 highlighted that students' curiosity of learning words has improved with the system, *"After a session student was curious about new words encountered."* Teachers noted that the system can be used in speech therapy activities, *"Students can pronounce displayed word following my lips, which is ideal as they get to see the actual word as well."* (T1).
Engaging Learning Experience: Teachers noted students' increased engagement in classroom sessions, *"System is interactive for them, especially Karaoke-Like presentations, they seem very engaged during classroom sessions."* (T2)
Enhanced Learning Speed and Effectiveness: Students and teachers noted the system's ability to improve their learning speed, helping them follow classroom sessions more effectively: *"Now, I don't have to look at the book to read the words."* (S1), T2 mentioned, *"I can verbally deliver the classroom session, we don't have to write everything in a whiteboard."*
Challenges and Concerns: Students reported challenges in the system, including the need for assistance with the device setup, discomfort due to its weight, eye strain, and dizziness after prolonged use.

6 CONCLUSION

In this work, we explored the effective utilization of a real-time AR caption interface to enhance the learning experience of DHH students in specialized educational settings. Our comprehensive need analysis identified the unique challenges in a School of Deaf educational setting that impact the learning experience of DHH students. Adopting a User-Centered Design process with ideal designs from Co-Design, we developed effective and interactive caption presentation methods, placement methods, and visualizations for transcription errors to enhance the overall learning experience. Strong user preference, higher usability scores, and qualitative feedback suggest that tailored AR real-time captioning interfaces can enhance the learning experience, support educational activities, and improve the literacy development of DHH students in specialized settings. However, our work is done within a single school of deaf with one-on-one interactions between a student and a teacher, and in future work, we intend to optimize our interfaces and expand our study to include more comprehensive evaluations involving a larger, diverse group of DHH students across different educational settings.

REFERENCES

[1] Nicoletta Adamo-Villani and Saikiran Anasingaraju. 2017. Holographic Signing Avatars for Deaf Education. In *E-Learning, E-Education, and Online Training*. Springer International Publishing, Cham, 54–61. https://doi.org/10.1007/978-3-319-49625-2_7

[2] Murat Akçayır and Gökçe Akçayır. 2017. Advantages and challenges associated with augmented reality for education: A systematic review of the literature. *Educational Research Review* 20 (2017), 1–11. https://doi.org/10.1016/j.edurev.2016.11.002

[3] Faruk Arici, Pelin Yildirim, Şeyma Caliklar, and Rabia M. Yilmaz. 2019. Research trends in the use of augmented reality in science education: Content and bibliometric mapping analysis. *Comput. Educ.* 142, C (dec 2019), 23 pages. https://doi.org/10.1016/j.compedu.2019.103647

[4] Stephanie Berger, Oliver Niebuhr, and Kerstin Fischer. 2018. Eliciting extra prominence in read-speech tasks: The effects of different text-highlighting methods on acoustic cues to perceived prominence. In *Proc. Speech Prosody 2018*. International Speech Communication Association (ISCA), Poznań, Poland, 75–79. https://doi.org/10.21437/SpeechProsody.2018-15

[5] Larwan Berke, Christopher Caulfield, and Matt Huenerfauth. 2017. Deaf and Hard-of-Hearing Perspectives on Imperfect Automatic Speech Recognition for Captioning One-on-One Meetings. In *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility* (Baltimore, Maryland, USA) *(ASSETS '17)*. Association for Computing Machinery, New York, NY, USA, 155–164. https://doi.org/10.1145/3132525.3132541

[6] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101. http://eprints.uwe.ac.uk/

11735

[7] John Brooke. 1995. SUS: A quick and dirty usability scale. *Usability Eval. Ind.* 189 (11 1995).

[8] Andy Brown, Rhia Jones, Mike Crabb, James Sandford, Matthew Brooks, Mike Armstrong, and Caroline Jay. 2015. Dynamic Subtitles: The User Experience. In *Proceedings of the ACM International Conference on Interactive Experiences for TV and Online Video* (Brussels, Belgium) *(TVX '15)*. Association for Computing Machinery, New York, NY, USA, 103–112. https://doi.org/10.1145/2745197.2745204

[9] Janine Butler, Brian Trager, and Byron Behm. 2019. Exploration of Automatic Speech Recognition for Deaf and Hard of Hearing Students in Higher Education Classes. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility* (Pittsburgh, PA, USA) *(ASSETS '19)*. Association for Computing Machinery, New York, NY, USA, 32–42. https://doi.org/10.1145/3308561.3353772

[10] Ed H. Chi, Lichan Hong, Michelle Gumbrecht, and Stuart K. Card. 2005. ScentHighlights: Highlighting Conceptually-Related Sentences during Reading. In *Proceedings of the 10th International Conference on Intelligent User Interfaces* (San Diego, California, USA) *(IUI '05)*. Association for Computing Machinery, New York, NY, USA, 272–274. https://doi.org/10.1145/1040830.1040895

[11] Robert L. Duchnicky and Paul A. Kolers. 1983. Readability of Text Scrolled on Visual Display Terminals as a Function of Window Size. *Human Factors* 25, 6 (1983), 683–692. https://doi.org/10.1177/001872088302500605

[12] Matthew W. G. Dye, Dara E. Baril, and D. Bavelier. 2007. Which aspects of visual attention are changed by deafness? The case of the Attentional Network Test. *Neuropsychologia* 45 (2007), 1801–1811. https://doi.org/10.1016/j.neuropsychologia.2006.12.019

[13] Kenneth Forster. 1970. Visual perception of rapidly presented word sequences of varying complexity. *Perception & Psychophysics* 8 (07 1970), 215–221. https://doi.org/10.3758/BF03210208

[14] Juan Garzón, Juan Pavón, and Silvia Baldiris. 2019. Systematic review and meta-analysis of augmented reality in educational settings. *Virtual Reality* 23 (2019), 447–459. https://doi.org/10.1007/s10055-019-00379-9

[15] Sandra G. Hart. 2006. Nasa-Task Load Index (NASA-TLX); 20 Years Later. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 50, 9 (2006), 904–908. https://doi.org/10.1177/154193120605000909

[16] Richang Hong, Meng Wang, Mengdi Xu, Shuicheng Yan, and Tat-Seng Chua. 2010. Dynamic captioning: video accessibility enhancement for hearing impairment. In *Proceedings of the 18th ACM International Conference on Multimedia* (Firenze, Italy) *(MM '10)*. Association for Computing Machinery, New York, NY, USA, 421–430. https://doi.org/10.1145/1873951.1874013

[17] Andri Ioannou and Vaso Constantinou. 2018. Augmented Reality Supporting Deaf Students in Mainstream Schools: Two Case Studies of Practical Utility of the Technology. In *Interactive Mobile Communication Technologies and Learning*. Springer International Publishing, Cham, 387–396. https://doi.org/10.1007/978-3-319-75175-7_39

[18] Dhruv Jain, Bonnie Chinh, Leah Findlater, Raja Kushalnagar, and Jon Froehlich. 2018. Exploring Augmented Reality Approaches to Real-Time Captioning: A Preliminary Autoethnographic Study. In *Proceedings of the 2018 ACM Conference Companion Publication on Designing Interactive Systems* (Hong Kong, China) *(DIS '18 Companion)*. Association for Computing Machinery, New York, NY, USA, 7–11. https://doi.org/10.1145/3197391.3205404

[19] Sushant Kafle and Matt Huenerfauth. 2016. Effect of Speech Recognition Errors on Text Understandability for People who are Deaf or Hard of Hearing. In *Proc. 7th Workshop on Speech and Language Processing for Assistive Technologies (SLPAT 2016)*. International Speech Communication Association (ISCA), San Francisco, USA, 20–25. https://doi.org/10.21437/SLPAT.2016-4

[20] Kellie Kercher and Dale C. Rowe. 2012. Improving the learning experience for the deaf through augment reality innovations. In *2012 18th International ICE Conference on Engineering, Technology and Innovation*. IEEE, Munich, Germany, 1–11. https://doi.org/10.1109/ICE.2012.6297673

[21] Raja Kushalnagar and Poorna Kushalnagar. 2014. Collaborative Gaze Cues and Replay for Deaf and Hard of Hearing Students. In *Computers Helping People with Special Needs*, Klaus Miesenberger, Deborah Fels, Dominique Archambault, Petr Peňáz, and Wolfgang Zagler (Eds.). Springer International Publishing, Cham, 415–422.

[22] Educational Publications Department Sri Lanka. 2020. *Sinhala Language and Literature - Grade 09*. Educational Publications Department, Sri Lanka, Kanduboda, Delgoda, Sri Lanka. http://www.edupub.gov.lk/Administrator/Sinhala/9/sin%20g-9/sinhala%20g9.pdf

[23] Greg Leigh and Kathryn Crowe. 2020. Evidence-Based Practices for Teaching Learners who are Deaf or Hard of Hearing in Regular Classrooms. https://doi.org/10.1093/acrefore/9780190264093.013.1258

[24] Shenzhen Haori Technology Co. Ltd. 2024. 2024 AR Smart Glasses 3D Augmented Reality Smartglasses Cardboard AR Headset AR Glasses. https://www.alibaba.com/product-detail/2022-AR-Smart-Glasses-3D-Augmented_60816962011.html?spm=a2700.shop_index.scGlobalHeaderSmall.51.6ce94b95LHWVCP

[25] John L. Luckner and Corey D. Pierce. 2013. Response to Intervention and Students Who are Deaf or Hard of Hearing. *Deafness & Education International* 15 (2013), 222 – 240. https://doi.org/10.1179/1557069X13Y.0000000027

[26] Le Luo, Dongdong Weng, Guo Songrui, Jie Hao, and Ziqi Tu. 2022. Avatar Interpreter: Improving Classroom Experiences for Deaf and Hard-of-Hearing People Based on Augmented Reality. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) *(CHI EA '22)*. Association for Computing Machinery, New York, NY, USA, Article 318, 5 pages. https://doi.org/10.1145/3491101.3519799

[27] Marc Marschark, Greg Leigh, Patricia Sapere, Denis Burnham, Carol Convertino, Michael Stinson, Harry Knoors, Mathijs P. J. Vervloed, and William Noble. 2006. Benefits of Sign Language Interpreting and Text Alternatives for Deaf Students' Classroom Learning. *The Journal of Deaf Studies and Deaf Education* 11, 4 (08 2006), 421–437. https://doi.org/10.1093/deafed/enl013 arXiv:https://academic.oup.com/jdsde/article-pdf/11/4/421/1146080/enl013.pdf

[28] Michael E. J. Masson. 1983. Conceptual processing of text during skimming and rapid sequential reading. *Memory & Cognition* 11, 3 (May 1983), 262–274. https://doi.org/10.3758/BF03196973

[29] Apostolos Meliones and Cosmin Duta. 2019. SeeSpeech: an Android application for the hearing impaired. In *Proceedings of the 12th ACM International Conference on PErvasive Technologies Related to Assistive Environments* (Rhodes, Greece) *(PETRA '19)*. Association for Computing Machinery, New York, NY, USA, 509–516. https://doi.org/10.1145/3316782.3324013

[30] Rhea Paul, Louise Paatsch, Naomi Caselli, Carrie Lou Garberoglio, Susan Goldin-Meadow, and Amy Lederberg. 2020. Current research in pragmatic language use among deaf and hard of hearing children. *Pediatrics* 146, Supplement_3 (2020), S237–S245.

[31] Yi-Hao Peng, Ming-Wei Hsi, Paul Taele, Ting-Yu Lin, Po-En Lai, Leon Hsu, Tzu-chuan Chen, Te-Yen Wu, Yu-An Chen, Hsien-Hui Tang, and Mike Y. Chen. 2018. SpeechBubbles: Enhancing Captioning Experiences for Deaf and Hard-of-Hearing People in Group Conversations. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (, Montreal QC, Canada,) *(CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–10. https://doi.org/10.1145/3173574.3173867

[32] Hector Ponce and Richard Mayer. 2014. An eye movement analysis of highlighting and graphic organizer study aids for learning from expository text. *Computers in Human Behavior* 41 (12 2014), 21–32. https://doi.org/10.1016/j.chb.2014.09.010

[33] Jason Proksch and D. Bavelier. 2002. Changes in the Spatial Distribution of Visual Attention after Early Deafness. *Journal of Cognitive Neuroscience* 14 (2002), 687–701. https://doi.org/10.1162/08989290260138591

[34] Luz Rello, Horacio Saggion, and Ricardo Baeza-Yates. 2014. Keyword Highlighting Improves Comprehension for People with Dyslexia. In *Proceedings of the 3rd Workshop on Predicting and Improving Text Readability for Target Reader Populations (PITR)*, Sandra Williams, Advaith Siddharthan, and Ani Nenkova (Eds.). Association for Computational Linguistics, Gothenburg, Sweden, 30–37. https://doi.org/10.3115/v1/W14-1204

[35] Rufat Rzayev, Paweł W. Woźniak, Tilman Dingler, and Niels Henze. 2018. Reading on Smart Glasses: The Effect of Text Position, Presentation Type and Walking. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) *(CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–9. https://doi.org/10.1145/3173574.3173619

[36] Matthew Seita, Sarah Andrew, and Matt Huenerfauth. 2021. Deaf and hard-of-hearing users' preferences for hearing speakers' behavior during technology-mediated in-person and remote conversations. In *Proceedings of the 18th International Web for All Conference* (Ljubljana, Slovenia) *(W4A '21)*. Association for Computing Machinery, New York, NY, USA, Article 25, 12 pages. https://doi.org/10.1145/3430263.3452430

[37] Matthew Seita, Sooyeon Lee, Sarah Andrew, Kristen Shinohara, and Matt Huenerfauth. 2022. Remotely Co-Designing Features for Communication Applications using Automatic Captioning with Deaf and Hearing Pairs. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) *(CHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 460, 13 pages. https://doi.org/10.1145/3491102.3501843

[38] Unity Technologies. 2023. https://unity.com/releases/editor/whats-new/2021.3.25

[39] Eric Urban. 2024. What is the Speech Service? - Azure AI services. https://learn.microsoft.com/en-us/azure/ai-services/speech-service/overview

[40] Thavindu Ushan. 2024. SeEar-SDK. https://github.com/ThavinduUshan/SeEar-SDK

[41] Keith Vertanen and Per Ola Kristensson. 2008. On the Benefits of Confidence Visualization in Speech Recognition. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) *(CHI '08)*. Association for Computing Machinery, New York, NY, USA, 1497–1500. https://doi.org/10.1145/1357054.1357288

[42] Fangzhou Wang, Hidehisa Nagano, Kunio Kashino, and Takeo Igarashi. 2016. Visualizing Video Sounds With Sound Word Animation to Enrich User Experience. *IEEE Transactions on Multimedia* PP (09 2016), 1–1. https://doi.org/10.1109/TMM.2016.2613641

[43] Wikipedia contributors. 2024. Sinhala language — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Sinhala_language&oldid=1211757750

[44] A.H. Ziadat and A.A. Al Rahmneh. 2020. The learning, social, and economic challenges facing the deaf and hearing-impaired individuals. *Cypriot Journal of Educational Sciences* 15, 5 (Oct. 2020), 976–988. https://doi.org/10.18844/cjes.v15i5.5130

A APPENDIX

A.1 System Design

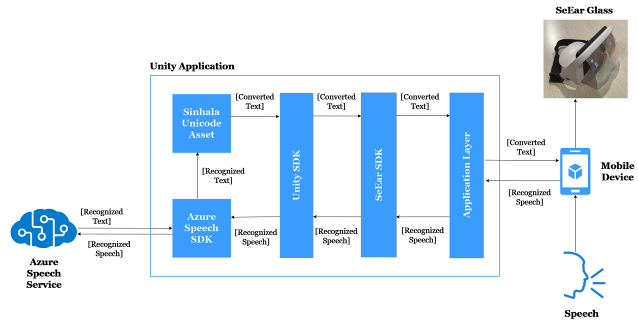


Figure 4: High-level architecture diagram